

NEURO OPHTHALMOLOGY

REQUIRED READINGS: Blumenfeld, Blumenfeld, Chapter 13

LEARNING OBJECTIVES: after studying this chapter, students should be able to:

1. Explain the different types of eye movements and the 3 levels of eye movement control
2. Explain the role of cranial nerves III, IV and VI in the control of conjugated gaze
3. Name the deficits produced by lesion of the CN III, IV and VI and describe the symptoms and signs produced by these deficits. Recognize and differentiate them.
4. Name, localize and explain the function of the brainstem centers involved in the coordination of horizontal, vertical and vergence gaze
5. Recognize and describe the deficits produced by lesions of the brainstem centers that control gaze
6. Indicate the location and function of the cortical centers concerned with the control of ocular movements
7. Indicate the commands centers that are important in the production of saccades and smooth pursuits
8. Recognize and describe the deficits produced by lesions of the cerebral areas that control ocular movements
9. Explain the pathways for light and accommodation reflexes

OVERVIEW

In this chapter, we will study the pathways for the control and execution of eye movements. Eye movement abnormalities, may signal pathology in the brainstem, along the trajectory of the cranial nerves involved in ocular movements, or in the cerebral cortex.

Please refer to Blumenfeld textbook: Chapter 13, Eye movements and pupillary control, on page 566. Use these notes as a guide to your textbook readings.

Use the following comments on the Power Point Presentation as a guide to the subject.

Objective # 1- EYE MOVEMENTS

Slide 3: Why do we have conjugate eye movements? Why our eyes move together all the time or most of the time?

The functions of the conjugate movements of the eyes are essentially:

- To move a peripheral image or object of interest onto the center of the visual field (fovea) and
- To keep this image there even if the image or the head are moving.

Eye Movements

OBJ. # 1

Conjugate Movements of the eyes

Move a peripheral image onto the FOVEA  Keep this image on the FOVEA even if the image is moving or the head is moving



Slide 4: To do that our eyes move in different planes.

- Horizontal eye movements
- Vertical eye movements
- Vergence eye movements

Our eyes will produce well defined and precise movements in those planes which could be classified into slow or fast eye movements. Let's name and define these movements as:

- Saccades
- Smooth pursuits
- Vergence Movements: Convergence / Divergence. They are non-conjugate eye movements

Saccades are rapid eye movements that have the function of bringing an object of interest into the center of the visual field. We can elicit these kinds of movements voluntarily when we want to move our eyes from one object to another, but they can also be part of reflex movements. They can be produced in the horizontal or in the vertical plane.

Smooth pursuits are slow, tracking movements of the eyes that have the purpose of maintaining a moving object of interest in the center of the visual field. They are considered involuntary movements because we need a moving target for the movement to be performed.

Vergence movements are non-conjugate movements. They involve either convergence when the eyes move towards a target nearby, or divergence when the eyes move towards an object farther away. Convergence is one of the three components that characterize the accommodation or near reflex that we will see later in this chapter.



OBJ. # 1

Types Of Movements

- Horizontal Movements
- Vertical Movements
- Vergence Movements

Eye Movement Components

- Saccades
- Smooth pursuits



Slide 5: Reflexive eye movements are used constantly and include the optokinetic reflex also sometimes called optokinetic nystagmus, and the vestibulo-ocular reflex.

Optokinetic nystagmus is a normal reflex elicited by visual fixation of a object that is moving in the visual field until it disappear. The eyes follow the object with a smooth pursuit movement and then a reflexive saccade fixates another object coming into view, and so on. A typical example will be following a series of trees when sitting in a car and looking through the window. This is not a conscious movement, it is a reflex.

The presence of a spontaneous nystagmus when the patient is looking straight ahead, and not moving the head, **is abnormal**.

The vestibulo-ocular reflex (VOR) is a reflex that stabilizes the eyes when fixating a stationary object and compensates for head movements. The reflex is triggered when you fixate an object of interest while your head moves in space away from the object. The eyes then move the same amount but in the opposite direction of the head movement to maintain the object of interest in the center of the visual field.

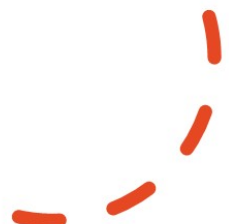
Notice that optokinetic nystagmus occurs when objects move in space while the head is stable, but the VOR occurs when the head moves while fixing an object of interest which is still.

Nystagmus is a complex eye movement with 2 components: a slow phase, smooth pursuit movement followed by a rapid phase, the saccade. Nystagmus is normal only when is elicited reflexively. Click on the picture to see an example of nystagmus.

OBJ. # 1

- Optokinetic reflex = Triggered by an object moving in the outside world
- Vestibulo-ocular reflex (VOR) = Stabilizes the eyes on a visual image during head and body movements.
- The VOR can be used to assess the integrity of the eye movement pathways in the brainstem in comatose patients
- NYSTAGMUS = a type of ocular movement with 2 characteristic phases:
 - a slow drift, followed by
 - a saccade in the opposite direction

Reflexive Eye Movements



Slide 6, 7: Contain links to eye movement example videos

LEVELS OF EYE MOVEMENT CONTROL

Slide 8: There are 3 levels of eye movement control:

- Control by the lower motor neurons (LMN): located in the nuclei of CN III, IV, and VI. The LMNs provide motor commands to the extrinsic or extraocular muscles of the eyes.
- In turn the LMNs are controlled by supranuclear centers in the brainstem: **the gaze centers**
- The ultimate level of control is provided by 2 areas in the cerebral cortex: **the frontal and parietal eye fields**



Control Of Eye Movements

OBJ. # 1

There are 3 levels of control

- The cranial nerve level of control
- The Gaze Centers level of control
- The cortex level of control

Objective # 2: CRANIAL NERVE LEVEL OF CONTROL

Slide 9: Brief reminder of the innervations of extrinsic and intrinsic eye muscles by:

- Oculomotor
- Trochlear
- Abducens

OBJ. # 2

Cranial Nerve Level Control

**Extrinsic eye muscles
(skeletal muscles)**

- Oculomotor nerve, CN III
- Abducens nerve, CN VI
- Trochlear nerve, CN IV

**Intrinsic eye muscles
(smooth muscles)**

- Pupillary constrictor and dilator
- Ciliary muscles

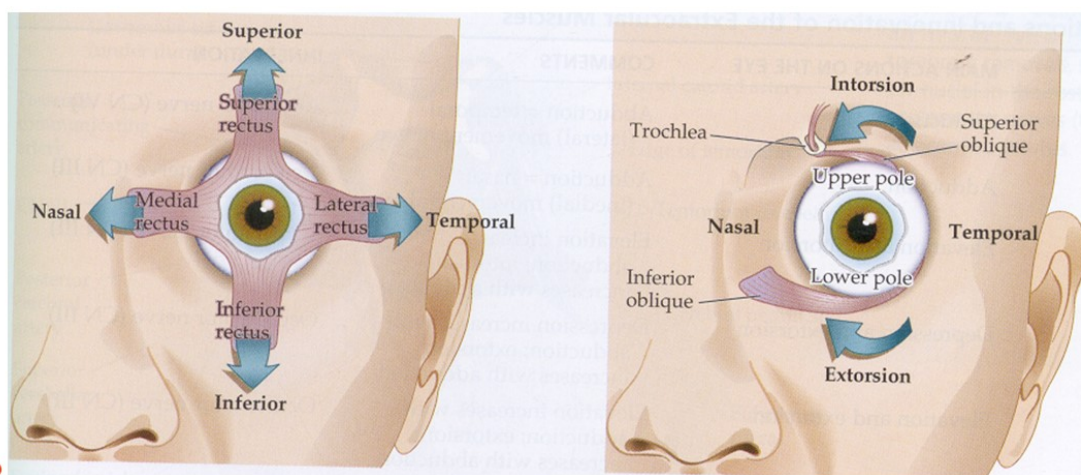


Autonomic innervation

Slide 10: Diagram taken from Blumenfeld showing the direction of pulling of extrinsic muscles of the eye

Extrinsic Eye Muscles

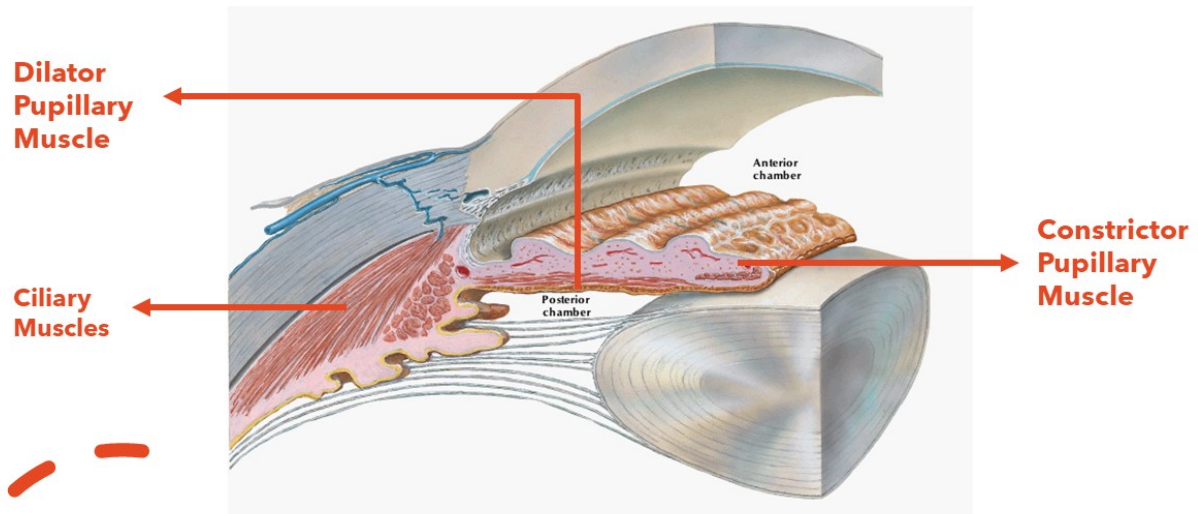
OBJ. # 2



Slide 11: Diagram showing the location of the intrinsic eye muscles: ciliary and constrictor/dilator pupillary muscles.

Intrinsic Eye Muscles

OBJ. # 2



Slide 12: Brief review of the function of cranial nerve III as well as the name and location of the nuclei associated with it.

Oculomotor, CN III

OBJ. # 2

Function

- Motor innervation of extrinsic muscles of the eye
- Parasympathetic motor innervation of smooth muscles of the eye

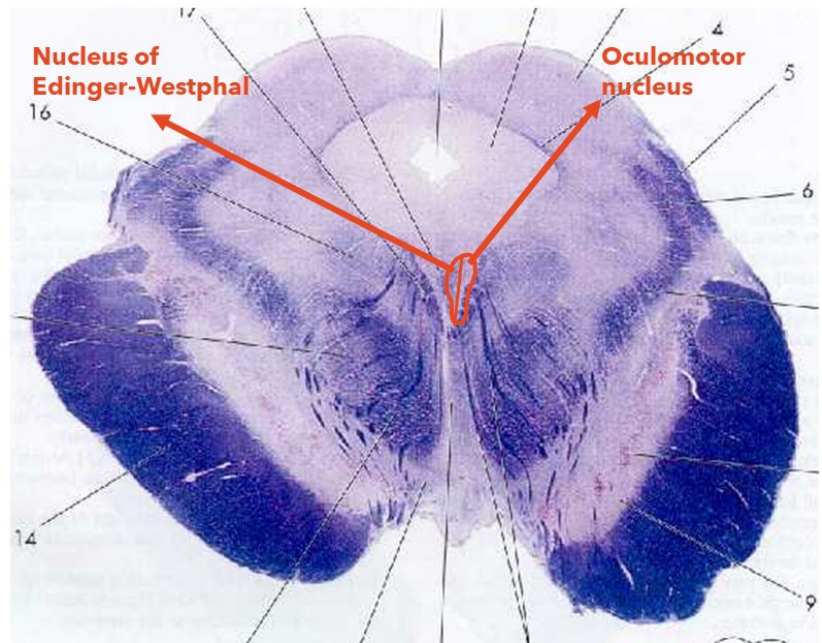
Nuclei

- Oculomotor Nucleus (rostral midbrain) → GSE
- Nucleus of Edinger - Westphal (parasympathetic, rostral midbrain) → GVE

Slide 13: Diagram of the rostral midbrain showing the location of oculomotor nucleus and the parasympathetic nucleus of Edinger-Westphal, both nuclei associated with CN III.

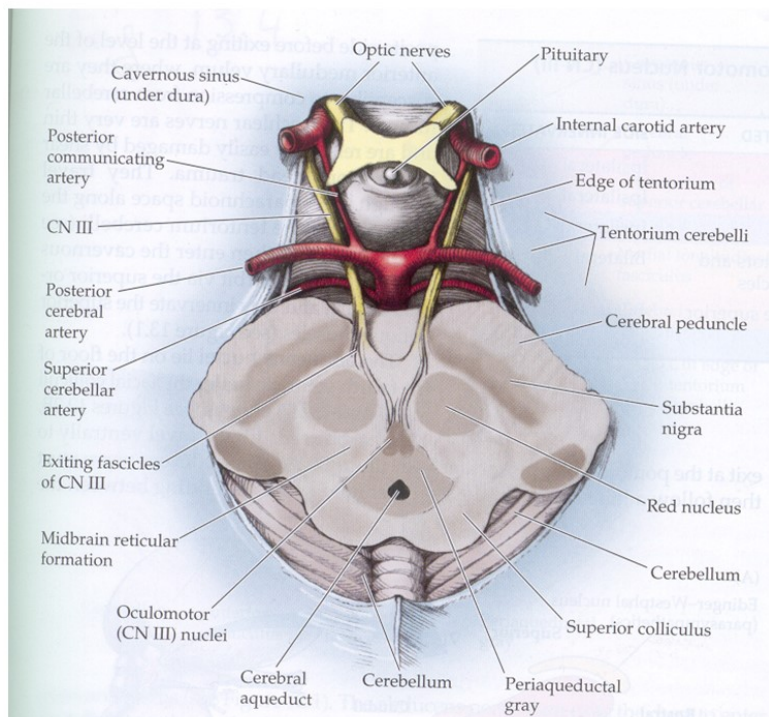
Oculomotor Nerve

OBJ. # 2



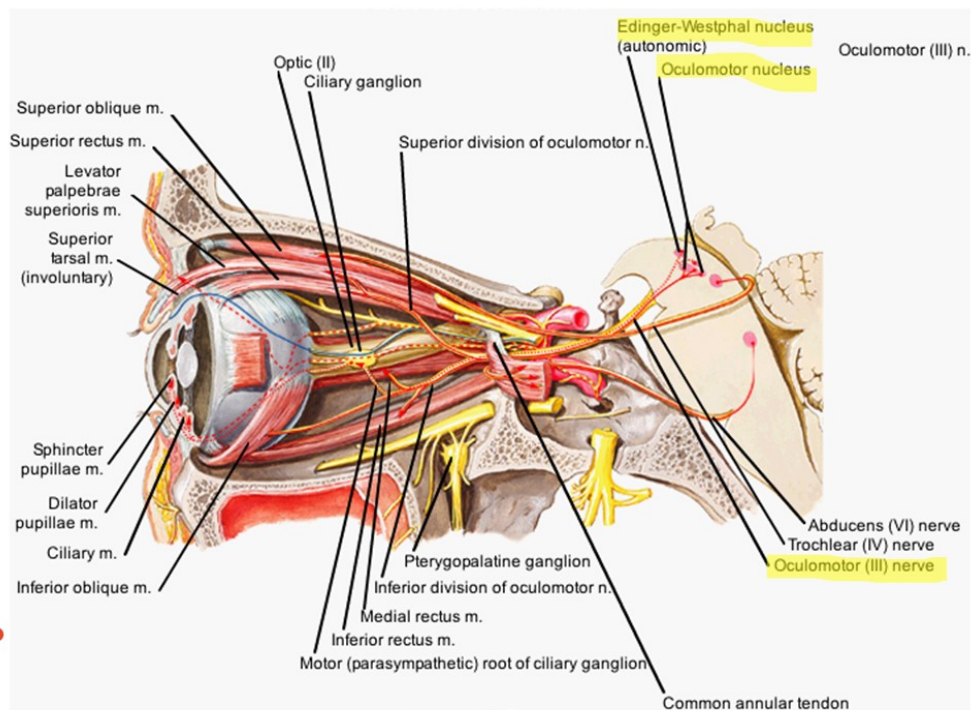
Slide 14: Diagram of the rostral midbrain showing the trajectory of oculomotor nerve in the subarachnoid space. Notice the relationship between the nerve and the arteries. The nerve exits between the 2 cerebral peduncles in an area known as the interpeduncular fossa and travels between the posterior cerebral artery superiorly and the superior cerebellar artery inferiorly. In the subarachnoid space, it travels parallel to the posterior communicating artery and very close to its bifurcation from the internal carotid artery. Any damage to these arteries such as the presence of a growing saccular aneurysm or the rupture of an aneurysm with bleeding can affect the function of oculomotor nerve, producing **oculomotor palsy**.

- **Clinical Note:** The autonomic fibers carried along CNIII travel in the periphery and tend to be more affected by compressive lesions, while ischemic lesions such as a diabetic third nerve palsy can spare the pupil.



OBJ. # 2

Slide 15: Diagram taken from Netter atlas showing the orbit, the extrinsic muscles, and the nerves innervating them.



OBJ. # 2

Slide 16: The oculomotor nerve provides the efferent limb of 2 clinically important reflexes: the light and accommodation reflexes.

OBJ. # 2

Oculomotor Nerve, CN III

Reflexes:

- Light and accommodation reflexes
 - Afferent limb: optic nerve
 - Efferent limb: oculomotor nerve

Slide 17: The trochlear nerve, CN IV, function and location of the nerve nucleus

OBJ. # 2

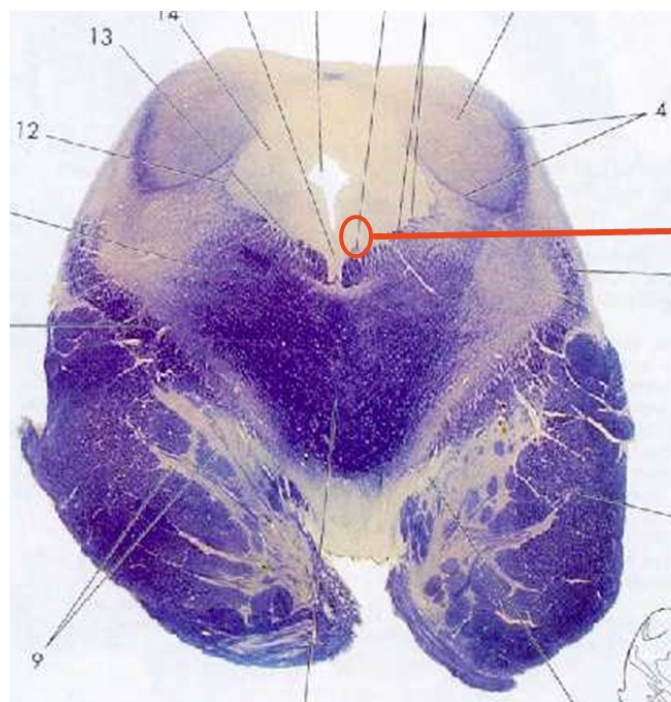
Trochlear Nerve, CN IV

Function → Motor innervation of superior oblique muscle of the eye

Nucleus → Trochlear nucleus(caudal midbrain) → GSE

Slide 18: Diagram showing a section through the caudal midbrain indicating the location of the trochlear nucleus.

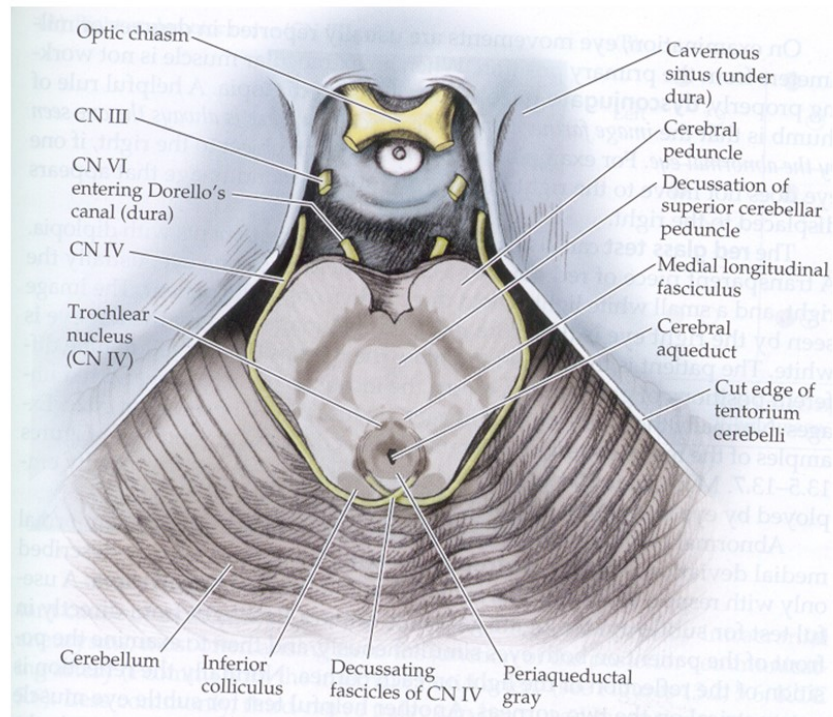
OBJ. # 2



**Trochlear
nucleus**

Slide 19: Diagram showing the trajectory of the trochlear nerve fibers inside the brainstem and in the subarachnoid space. Notice that the nerve fibers originate from the nucleus located in the anterior periaqueductal gray matter, they embrace the periaqueductal gray and cross posteriorly before exiting the brainstem.

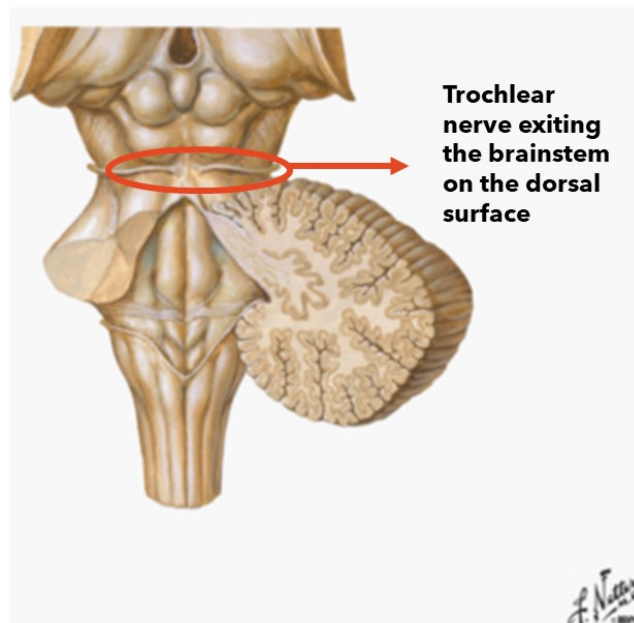
OBJ. # 2



Slide 20: Dorsal surface of the brainstem showing the point where the trochlear nerve exits dorsally.

Anatomy of the Fourth Ventricle – Posterior view

OBJ. # 2



Slide 21: The abducens nerve, CN VI - function and location of its nucleus in the brainstem

OBJ. # 2

Abducens Nerve, CN VI

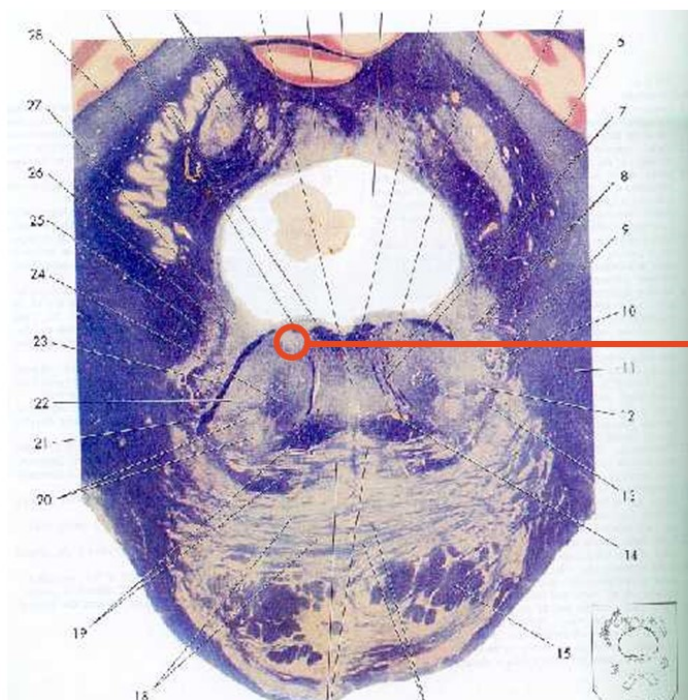
Function → Motor innervation of the lateral rectus muscle of the eye

Nucleus → Abducens nucleus(caudal pons) → GSE



Slide 22: Diagram showing the abducens nucleus in the caudal pons section

OBJ. # 2



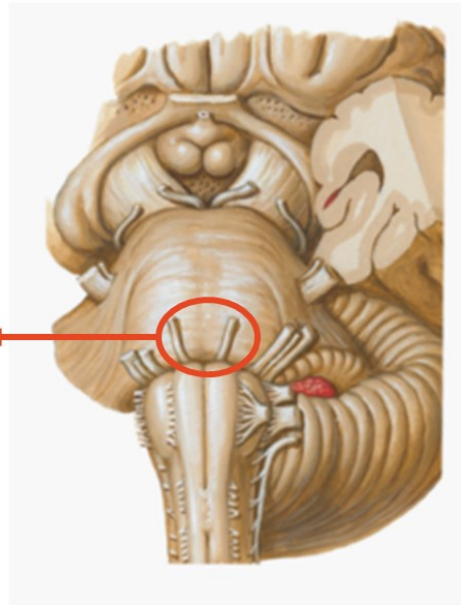
Abducens nucleus



Slide 23: Anterior or ventral surface of the brainstem showing the cranial nerves, in particular abducens nerve.

OBJ. # 2

Abducens Nerve



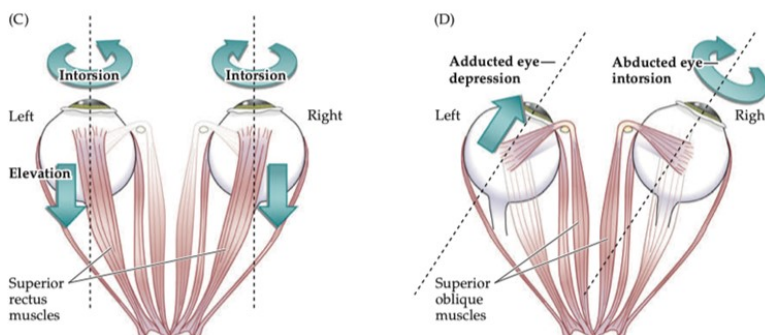
**Brain Stem
Surface
Anatomy:
Anterior View**

Objective # 3 – Deficits produced by lesions of the cranial nerve nuclei or their peripheral fibers

Slide 24: Diagram showing the eye movements performed by oculomotor nerve.

Oculomotor Nerve

OBJ. # 3



MUSCLE

Medial rectus
Superior rectus
Inferior rectus
Inferior oblique

MAIN ACTION

Adduction
Elevation and intorsion
Depression and extorsion
Elevation and extorsion

INNERVATION

Oculomotor nerve, CN III
Oculomotor nerve, CN III
Oculomotor nerve, CN III
Oculomotor nerve, CN III

Slide 25: This slide describes **oculomotor palsy**, the deficit produced by a lesion of the oculomotor nerve fibers or its nucleus in the brainstem. Patients present, as can be observed in **Slide 26 and 27**, with the affected eye in a ‘down and out’ position when the eyes are in primary gaze. The patient will experience **diplopia** (double vision) mostly when attempting gaze towards the nose and up. There is also **ptosis** of the affected eyelid due to the paralysis of the levator palpebrae superioris (**Slide 27**). Involvement of the parasympathetic fibers from the nucleus of Edinger-Westphal produces **mydriasis**, paralytic dilation of the affected pupil. The pupils are then unequal in size which is called **anisocoria**. Please refer to Blumenfeld textbook, Key Clinical Concept 13.2, on page 572.

Lesion of CN III or its nucleus produces paralysis of all muscles innervated by it. The only movements left are:

- Some abduction
- Some depression and intorsion



- The eye is in a **Down And Out** position at rest
 - DIPLOPIA = Double vision
 - Ptosis = Paralysis of the levator palpebrae superioris
- If there is involvement of parasympathetic fibers



- Anisocoria = Pupillary asymmetry and
- Mydriasis = unresponsive dilated pupil

OBJ. # 3

Oculomotor Palsy

Oculomotor Nerve Deficits

Slide 26, 27 show some examples.



OBJ. # 3

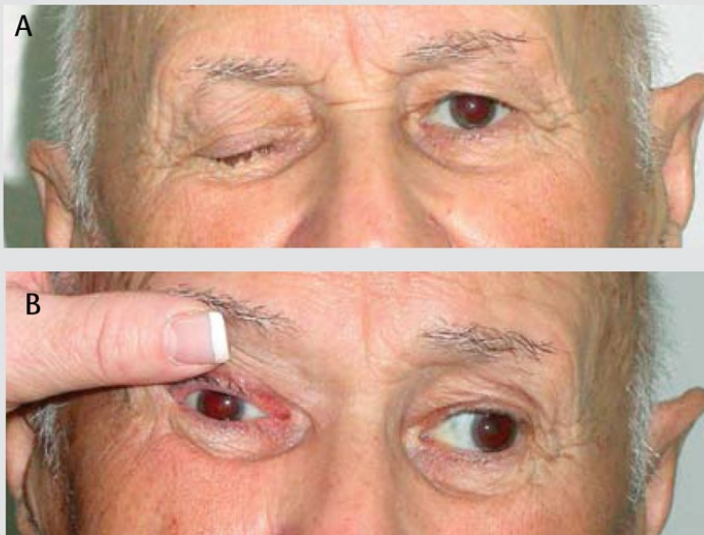
The R eye is in a down and out position

Oculomotor Palsy

**Symptoms
consistent
with palsy of
third cranial
nerve**

OBJ. # 3

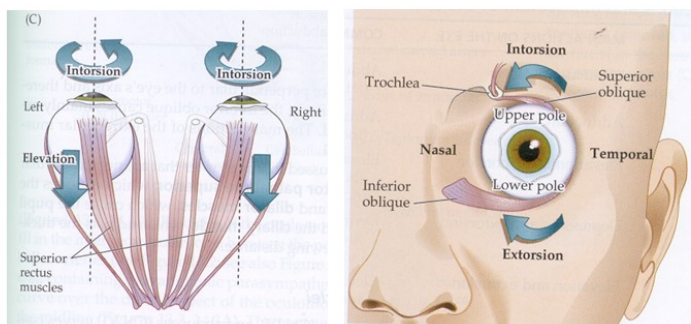
A) Complete right upper lid ptosis in primary position and B) complete limitation of adduction of the right eye on left gaze.



Slide 28: Diagram showing the function of trochlear nerve, CN IV

Trochlear Nerve

OBJ. # 3



MUSCLE

Superior oblique

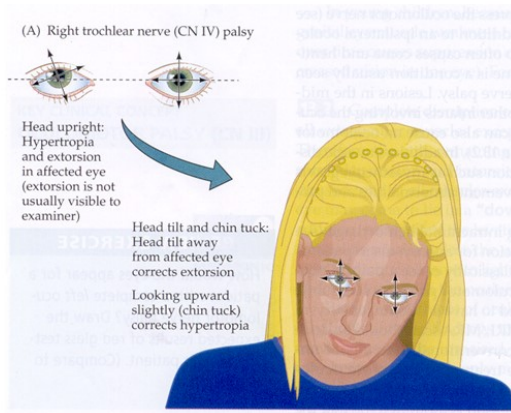
MAIN ACTION

Depression and
intorsion

INNERVATION

Trochlear nerve, CN IV

Slide 29: This slide describes the symptoms and signs produced by **trochlear palsy**. The patient has **extorsion** of the affected eye with the pupil in a higher position in the orbit compared with the normal eye. This is called **hypertropia** and is produced by the unopposed action of the inferior oblique muscle. The patient complains of vertical diplopia when gazing down and they tend to look upward (with eyes, downward with head) and tilt the head away from the affected side to correct for the extorsion and the uneven location of the eyes in the orbit.



Trochlear Palsy

Trochlear Nerve Deficits

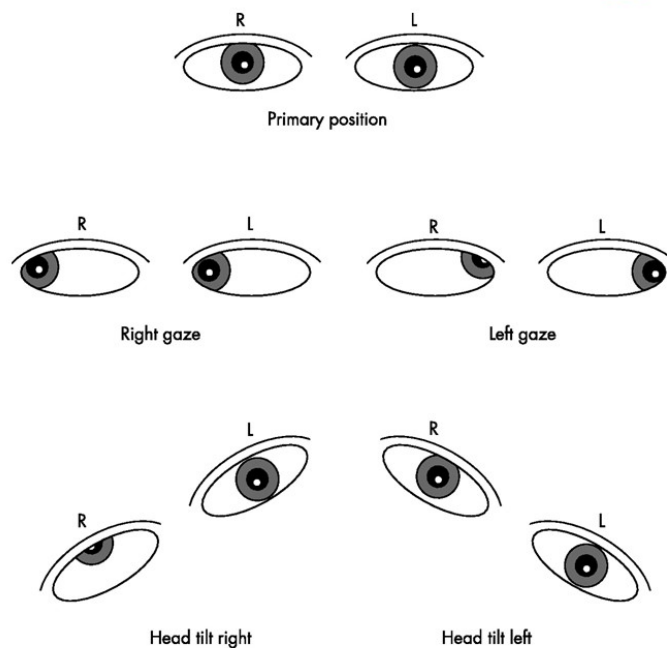
- Extorsion of the eye
- Hypertropia = The eye is in a higher position at rest
- Vertical diplopia that worsens when the patient looks down and toward the nose
- Patients tend to look up and tilt the head away from the affected eye

OBJ. # 3

Slide 30: Diagram showing the position of the eyes in the orbit on R or L gaze and head tilting. This helps to explain the patient symptoms.

OBJ. # 3

Trochlear Palsy



Slide 31 shows an adult patient with trochlear palsy while **slide 31** shows a child with a very evident defect.



OBJ. # 3

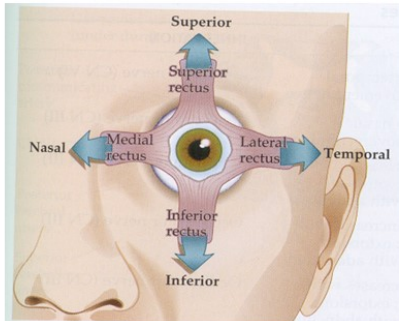
Trochlear Palsy

Slide 33: Diagram showing the function of abducens nerve.



OBJ. # 3

Abducens Nerve



MUSCLE

Lateral rectus

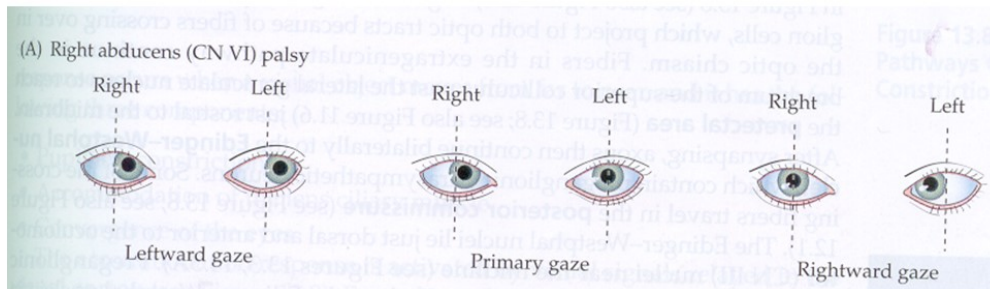
MAIN ACTION

Abduction

INNERVATION

Abducens nerve, CN VI

Slide 34: This slide describes the symptoms and signs of **abducens palsy** which occurs when the **abducens nerve fibers** are damaged. Damage of the abducens nucleus in the pons produces a different deficit that will be addressed later. Abducens palsy produces an abnormal medial deviation of the affected eye or esotropia on primary gaze, and the patient experiences horizontal diplopia. When the patient tries to gaze to the affected side, we can see how the affected eye does not move farther than the midline.



A lesion of the abducens nerve fibers will produce:

- Horizontal diplopia
- Esotropia of the affected eye = abnormal medial deviation of the eye
- Diplopia worsens with abduction of affected eye

Some Patients tend to turn the head toward the affected eye

OBJ. # 3

Abducens Palsy

Abducens Nerve Deficits

Slide 35, 36 are examples of abducens palsy. In slide 35 you can see the patient's impairment in primary gaze and when attempting gaze to the right and to the left.

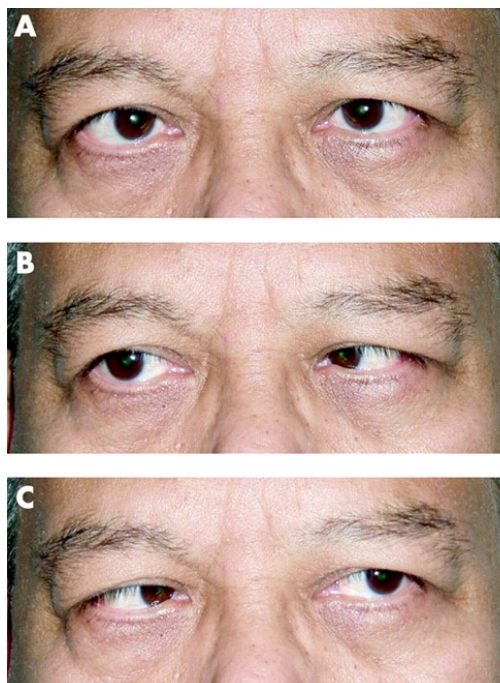


OBJ. # 3

Affected eye is medially deviated
in primary position

Abducens Palsy

Abducens Palsy



Objective 4: SUPRANUCLEAR CONTROL OF EYE MOVEMENTS

Slide 38, 39: The brainstem controls the direction of eye movements through three gaze centers: **the horizontal gaze center, the vertical gaze center and the vergence gaze center.** They are responsible for the coordination of both eyes during conjugate movement. These centers project to the appropriate cranial nerve nuclei to coordinate muscle activation.

The horizontal gaze center is in the pontine reticular formation of the caudal pons near the abducens nucleus. It is named the **PPRF (paramedian pontine reticular formation)**. It consists of a collection of local circuit neurons that make decisions regarding which muscles need to contract or to relax to produce any conjugate horizontal movement of the eyes. This information is transmitted to the cranial nerve nuclei involved in horizontal motion **through the abducens nucleus.** The abducens nucleus is a central player in horizontal gaze because it receives information from the PPRF and distributes the information to directly controlling the ipsilateral lateral rectus muscle and communicates with the oculomotor nuclei on both sides to inform about which muscles need to be activated and which to be inhibited for a conjugate horizontal movement to occur.

The superior colliculus, the cerebellum and the vestibular nuclei are also important areas for controlling eye movements through projections to the gaze centers.

The horizontal gaze center in the pons also controls the accuracy of saccadic movements of the eyes in the horizontal plane.

Communication between gaze centers and cranial nerve nuclei is through nerve fibers ascending and descending in the medial longitudinal fasciculus (MLF).

Supranuclear Control Of Eye Movements

Brainstem control

OBJ. # 4

Horizontal gaze center

Para-median Pontine Reticular Formation (PPRF) + Abducens nucleus

Vertical and vergence gaze centers

Other control centers

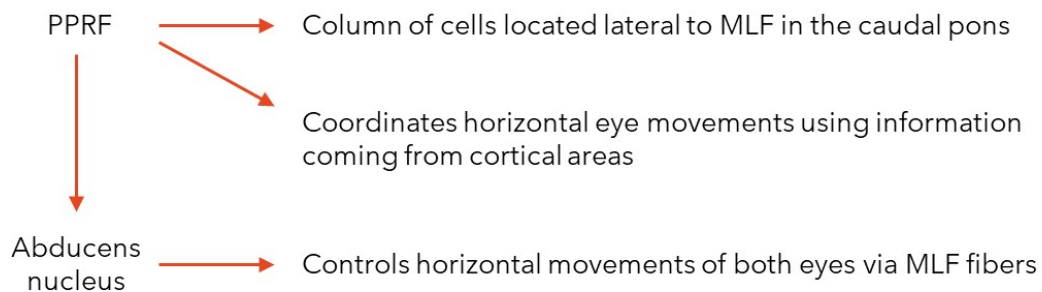
Midbrain reticular formation

- Superior colliculus
- Cerebellum
- Vestibular nuclei

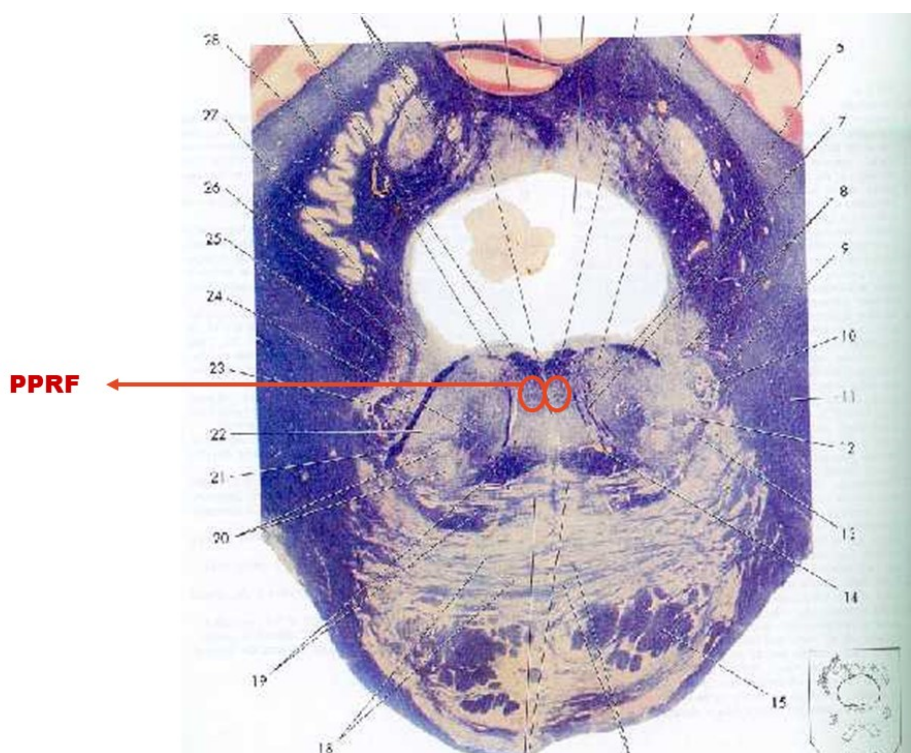
The Gaze Centers

Horizontal gaze centers

OBJ. # 4



Slide 40 is a section through the caudal pons showing the location of the PPRF.



OBJ. # 4

Slide 41: The vertical and vergence gaze centers are in the **reticular formation of the rostral midbrain**. The **vertical gaze center** consists of a group of neurons near the periaqueductal gray matter: a more dorsal area that regulates upgaze, and a more ventral area that regulates downgaze. An important nucleus in the ventral area is the **rostral interstitial nucleus of the MLF (RiMLF)**. This nucleus oversees coordination of vertical saccades also working with the superior colliculus. Vertical saccades are integrated bilaterally and communication between the two sides of the reticular formation is through the posterior commissure.

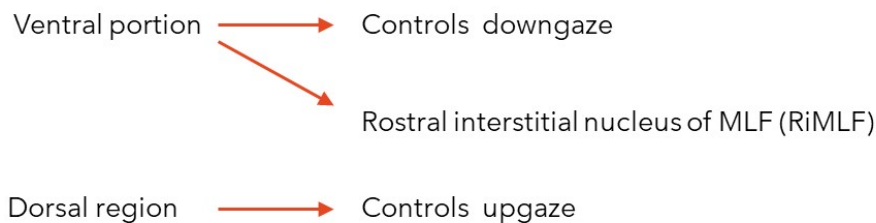
Vergence movements are under the control of visual pathways descending from the occipital and parietal cortices to the midbrain reticular formation. Even if the exact midbrain centers for the control of vergence movements have not been identified, it is believed that two separate groups of neurons in the midbrain reticular formation near the oculomotor nucleus coordinate convergence and divergence. These neurons are part of a nucleus named the interstitial nucleus of Cajal. Convergence movements are part of the accommodation reflex that we will study at the end of this chapter.

The Gaze Centers

OBJ. # 4

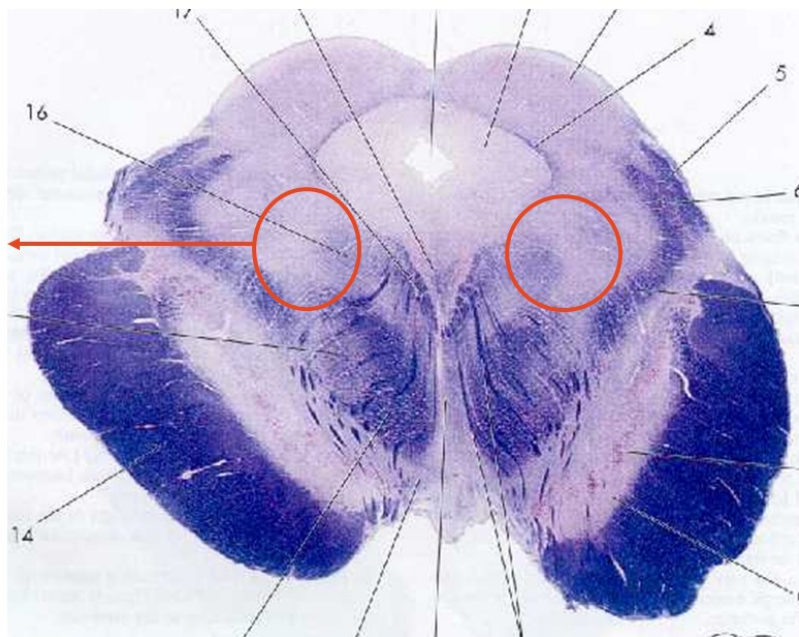
The rostral midbrain reticular formation controls vertical and vergence eye movements

Vertical gaze center



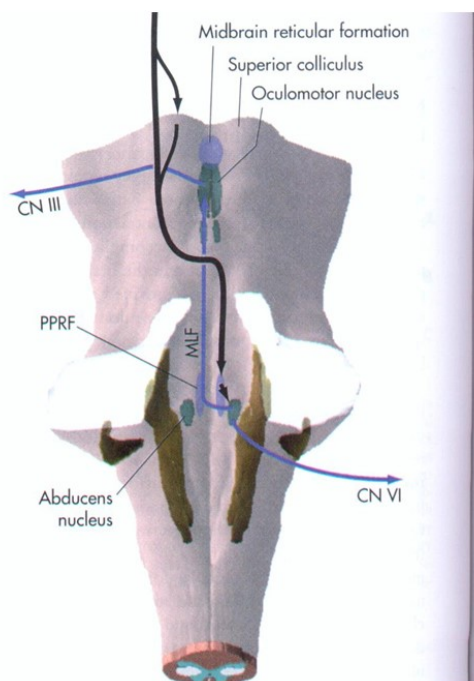
Slide 42 is a section of the rostral midbrain showing the location of the reticular formation where vertical and vergence gaze centers are found.

**Vertical and
Vergence gaze
centers**



OBJ. # 4

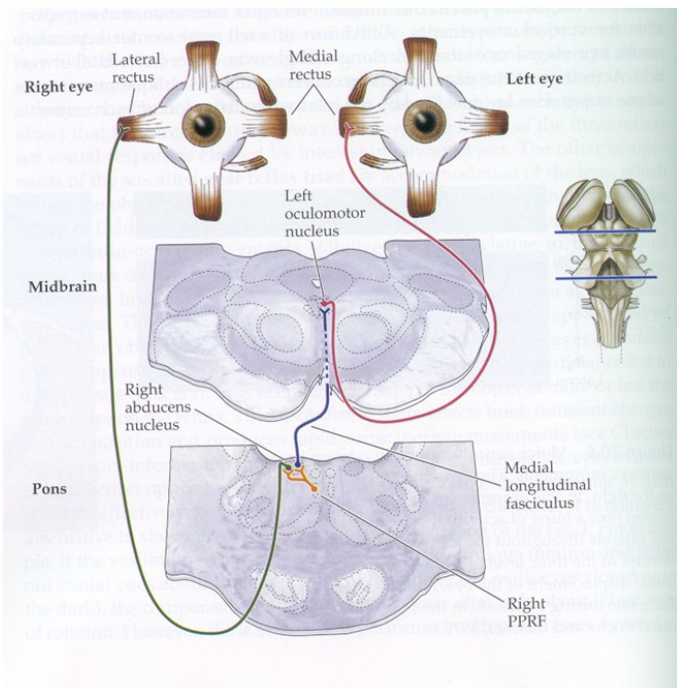
Slide 43: Diagram of the dorsal aspect of the brainstem showing the location of abducens nucleus and PPRF in the pons, oculomotor nucleus, and midbrain reticular formation. Ascending fibers in the MLF communicate the pons nuclei with the midbrain nuclei.



OBJ. # 4

Slide 44: This slide explains the control of horizontal gaze. The PPRF in the caudal pons receives a command from the cerebral cortex indicating the sense of the desired eye movement; let's say a saccade to the right, as in the picture. The right PPRF makes decisions regarding which muscles need to be activated or inhibited on both sides. The information is sent to the right abducens nucleus. There are 2 types of neurons in the abducens nucleus: the lower motor neurons which activate the lateral rectus muscle (green fibers in the diagram) and a group of neurons called internuclear neurons which transmit information to the contralateral oculomotor nucleus in the midbrain (blue fibers).

Moving the eyes to the right is performed by contracting the right lateral rectus and the left medial rectus. Therefore, the right abducens nucleus activates the right lateral rectus and sends excitatory signals to the left oculomotor nucleus to activate the left medial rectus. This internuclear communication is through the **MLF**. There is also an inhibitory pathway not shown in the diagram. The purpose of the inhibitory pathway is to prevent the simultaneous contraction of the antagonist muscles, in this case the left lateral rectus and the right medial rectus.



OBJ. # 4

**Diagram showing
the control of
horizontal gaze**

Objective # 5 HORIZONTAL GAZE DEFICITS PRODUCED BY BRAINSTEM DAMAGE

Slide 45: It is very important to differentiate between a lesion of the abducens fibers in the periphery (**abducens palsy**) from a lesion of the **abducens nucleus** in the caudal pons. In the case of abducens palsy, only the ipsilateral eye is involved while the contralateral eye moves in all directions. The only movement impaired is abduction of the affected eye. When a lesion of the **abducens nucleus** occurs, the situation is different because the abducens nucleus controls not only the ipsilateral lateral rectus but also communicates with oculomotor nucleus regarding the movement of the contralateral eye. Therefore, damage to the abducens nucleus impairs ipsilateral gaze; it affects the movement of both eyes to the side of the lesion. For this reason, the deficit is named **lateral gaze palsy**. When attempting gaze away from the lesion side the patient will not experience any deficit.

Brainstem Lesions Affecting Horizontal Gaze

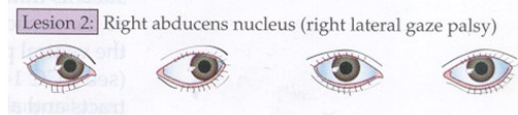
OBJ. # 5

Lesions of the abducens nerve: ABDUCENS PALSY

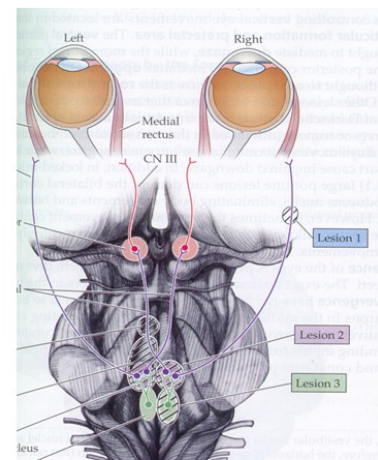


Abducens Palsy = one eye affected

Lesions of the abducens nucleus or PPRF- lateral gaze palsy



Lateral gaze palsy = both eyes affected



Slide 46: Internuclear ophthalmoplegia occurs due to damage of the **medial longitudinal fasciculus** in the pons. Therefore, the communication between abducens nucleus and the contralateral oculomotor nucleus is interrupted. The example in the diagram shows a left sided lesion. In this case when the patient looks to the left there is no deficit, left abducens nucleus and right MLF are functional. When the patient attempts gaze to the right, the right abducens nucleus tries to communicate with the left oculomotor nucleus, but the damage of the left MLF fibers makes this communication impossible. The right eye moves to the right, but the left eye stays in primary gaze. The right eye then moves back to the center, a movement that is called pseudo-nystagmus. This occurs likely to compensate for the diplopia that appears in this situation. In internuclear ophthalmoplegia vertical smooth pursuits and vestibulo-ocular reflex will be impaired because the vestibular pathways that control pursuits and VOR are no longer available. We will explain this last point later.

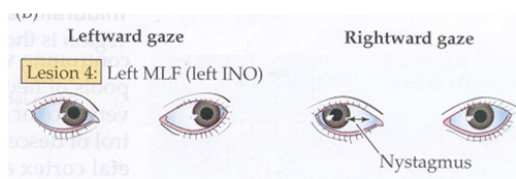
In **one and a half syndrome** the damage is more extensive and includes the **MLF fibers** and the **abducens nucleus** on the same side. The deficit produced is a combination of lateral gaze palsy (damage to abducens nucleus) when attempting gaze to the lesion side, and internuclear ophthalmoplegia (damage to MLF fibers) when attempting gaze away from the lesion side. The only eye movement left in these patients is the abduction of the eye on the unaffected side. Again, that eye will present pseudo-nystagmus trying to avoid diplopia.

In **'eight and a half syndrome'**, all the abnormalities of the eye movements from the one and half syndrome are included, but the ipsilateral facial nerve (CNVII) is affected as well. Don't read too much into this naming convention.

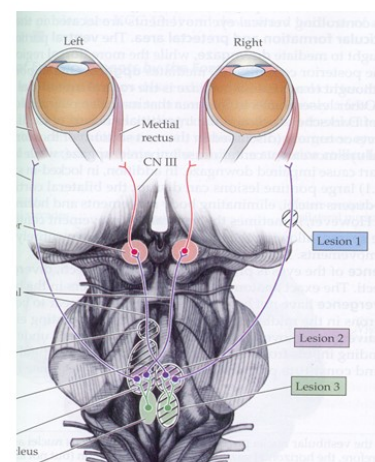
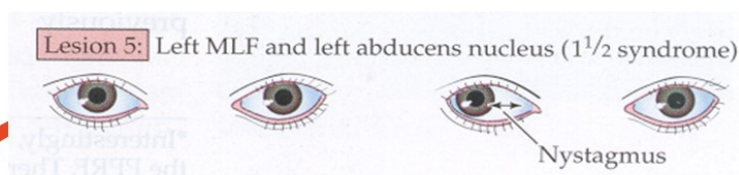
Brainstem Lesions Affecting Horizontal Gaze

OBJ. # 5

Lesions of the MLF - Internuclear Ophthalmoplegia



Lesions of the MLF + abducens nucleus ipsilaterally: one-and-a-half syndrome



Slide 47, 48, 49 – Some examples of these syndromes. Go to the Power Point presentation to see the videos.

Lesions affecting vertical gaze

Slide 50: Parinaud's syndrome is produced by compression of the dorsal midbrain and pretectal areas as occur in pineal gland tumors. There are 4 components of this syndrome (see Key Clinical concept 13.9 in Blumenfeld, page 588):

- Impaired vertical gaze.
- Large irregular pupil, non-reactive to light but reactive to near-to-far accommodation. This event is known as light-near dissociation.
- Eyelid abnormalities such as bilateral retraction or sometimes bilateral ptosis.
- Impaired convergence of the eyes

Brainstem Lesions Affecting Vertical Gaze

OBJ. # 5

Parinaud's Syndrome - Produced by a compression of the dorsal midbrain affecting the vertical and vergence gaze centers

Main symptoms are:

- Impair vertical gaze
- Pupillary light-near dissociation
 - Pupils unresponsive to light, but constrict with convergence
- Convergence-retraction nystagmus
- Collier's sign - lid retraction



Objective 6: FOREBRAIN CONTROL OF EYE MOVEMENTS

Slide 50: There are two cortical areas that have been identified as participating in the higher control of brainstem circuits. These are: the frontal eye fields and the parietal eye field.

The frontal eye field is located in the frontal lobe, close to the face motor area, and is the generator of saccades towards the opposite direction. Therefore, one side of the brain controls saccades to the contralateral side.

The parietal eye field is located at the confluence of parietal, occipital and temporal lobes. This area is the principal region in charge of smooth pursuit movements to the ipsilateral side.

Both areas are heavily influenced by visual inputs coming from the occipital lobe and are in constant communication with each other. The basal ganglia also play an important role in the modulatory control of eye movements.

Supranuclear Control Of Eye Movements

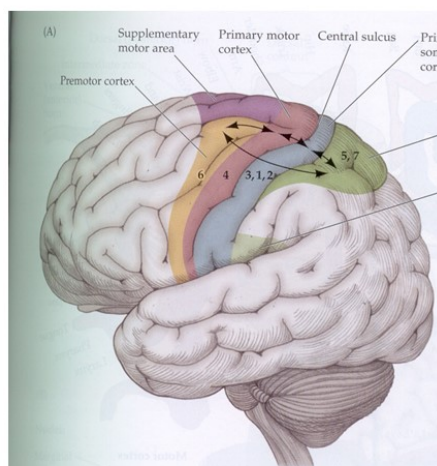
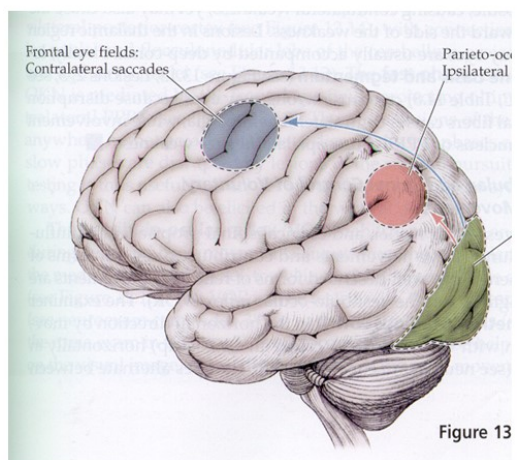


OBJ. # 6

- Cortical areas:
 - Frontal eye fields
 - Parieto-occipito-temporal cortex



Slide 52: Shows where these areas are located on the cerebral cortex.



OBJ. # 6

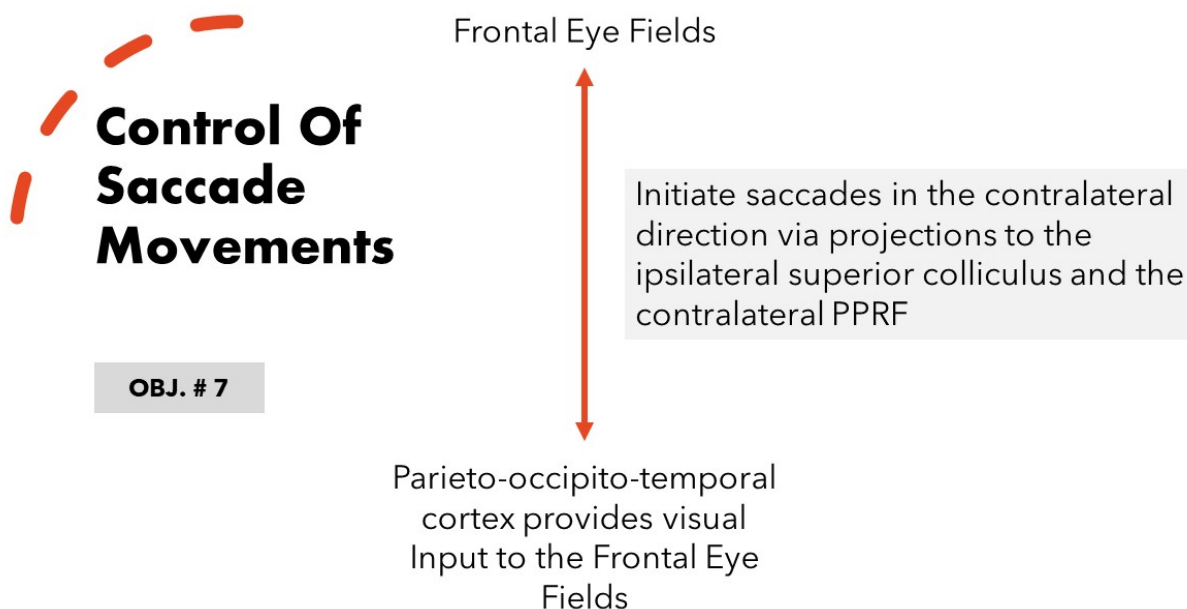
Frontal And Parietal Eye Fields

Objective # 7: CONTROL OF SACCADDES

Slide 53: As you remember from the beginning of this lecture the basic component of any eye movements are saccades and smooth pursuits. The commands necessary to produce these movements are mainly generated by the cerebral cortex.

To produce a saccade, the commands originate from the frontal eye field and are simultaneously sent to the ipsilateral superior colliculus and the contralateral horizontal or vertical gaze centers (PPRF/ midbrain reticular formation). The gaze centers perform the calculations regarding velocity and amplitude of the eye movement needed to reach a target relative to the present location of the eyes in space, and the location of the object of interest. The superior colliculus is a major player in the execution of saccades. The superior colliculus is a layered structure. The most superficial layer contains a map of the visual world and receives direct input from the retina. In this way, visual stimuli can elicit a reflexive saccade independently from the frontal eye fields by activation of neurons in the superficial layer of the colliculus. Saccades induced this way are known as “**express saccades**”.

In the case of horizontal saccades PPRF communicates with abducens nucleus for the distribution of commands to be performed by the cranial nerves involved. All these communications are via pathways traveling with MLF.

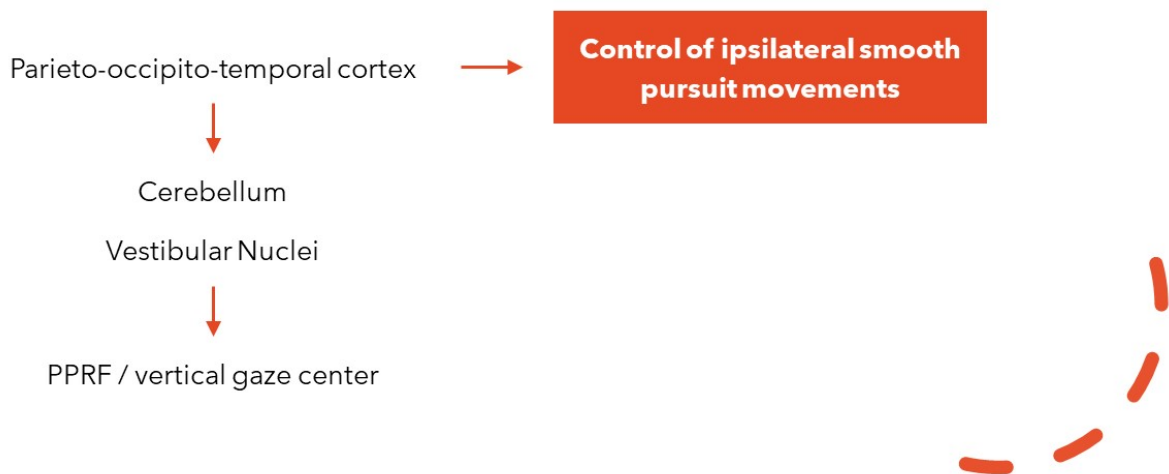


CONTROL OF SMOOTH PURSUITS

Slide 54: Commands for smooth pursuit movements originate in the parietal eye fields located in the confluence of the parietal-occipital-temporal cortex. These areas receive visual input from the occipital cortex. The parietal eye fields, and indirectly the frontal eye fields, send information to the cerebellum (flocculonodular lobe) and vestibular nuclei through pathways which are not well understood but are believed to follow the corticopontocerebellar pathways. The cerebellum and vestibular nuclei provide the final input to the gaze centers and abducens nucleus for the movement to be produced to the ipsilateral side.

Control Of Smooth Pursuit Movements

OBJ. #7



Objective # 8: Lesions affecting the cerebral cortex

Slide 55: Lesions affecting the frontal eye fields produce impaired saccadic eye movements to the contralateral side. In these patients there is a **gaze preference towards the side of the lesion** which is accompanied by weakness of the contralateral side of the body and face, due to involvement of the corticospinal and corticobulbar tracts.

A lesion of the parietal eye fields produces inability to generate smooth pursuits to the side of the lesion. The smooth pursuit deficit could be tested using the optokinetic reflex horizontally or vertically. A deficit of smooth pursuit movements determines that tracking an object in space is done using small saccades instead of a smooth movement of the eyes.

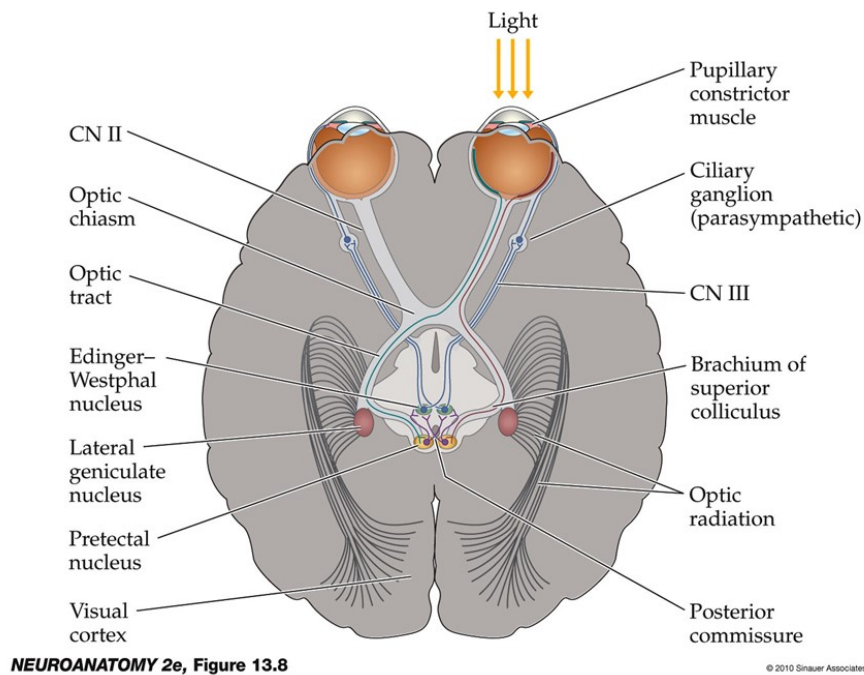
Eye Movement deficits due to cerebral cortex damage

OBJ. # 8

- Damage to the frontal eye fields in the motor cortex produces:
 - Impaired saccades to the contralateral side
 - “Gaze preference” towards the side of lesion
 - Usually there is weakness or hemiplegia on the contralateral side of the body and face
- Damage to the parietal eye fields produces:
 - Impaired smooth pursuits to the ipsilateral side

Objective # 9: Light and accommodation reflexes

Slide 56: Diagram of the light reflex pathway. Please see Blumenfeld Chapter 13, Eye movements and pupillary control, on page 566.



OBJ. # 9

**Light
Reflex**